Day 6: ML Models

- Linear Regression
- Logistic Regression
- Decision Trees

Introduction to Core ML Algorithms

What are ML Models?

- Algorithms that learn patterns from data
- Used to make predictions or classifications
- Core building blocks of AI systems

Linear Regression: Predicting Continuous Values

What it does:

- Predicts numerical values
- Finds relationship between variables

Example:

- House price prediction
- Salary prediction
- Sales forecasting

Key idea:

- Fits a straight line to data

Logistic Regression: Classification Model

What it does:

Predicts categories (not numbers)

Outputs probability

Example:

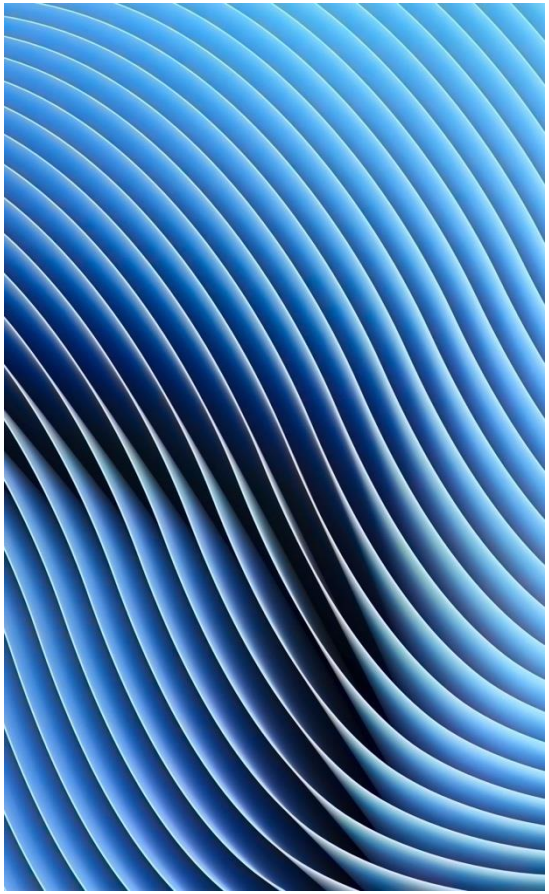
Spam vs not spam

Yes or No decisions

Pass or fail prediction

Key idea:

Uses sigmoid function to classify results



Decision Trees: Rule-Based Learning Model

What it does:

- Splits data into branches (like a tree)
- Makes decisions step-by-step

Example:

- Loan approval
- Customer classification
- Medical diagnosis

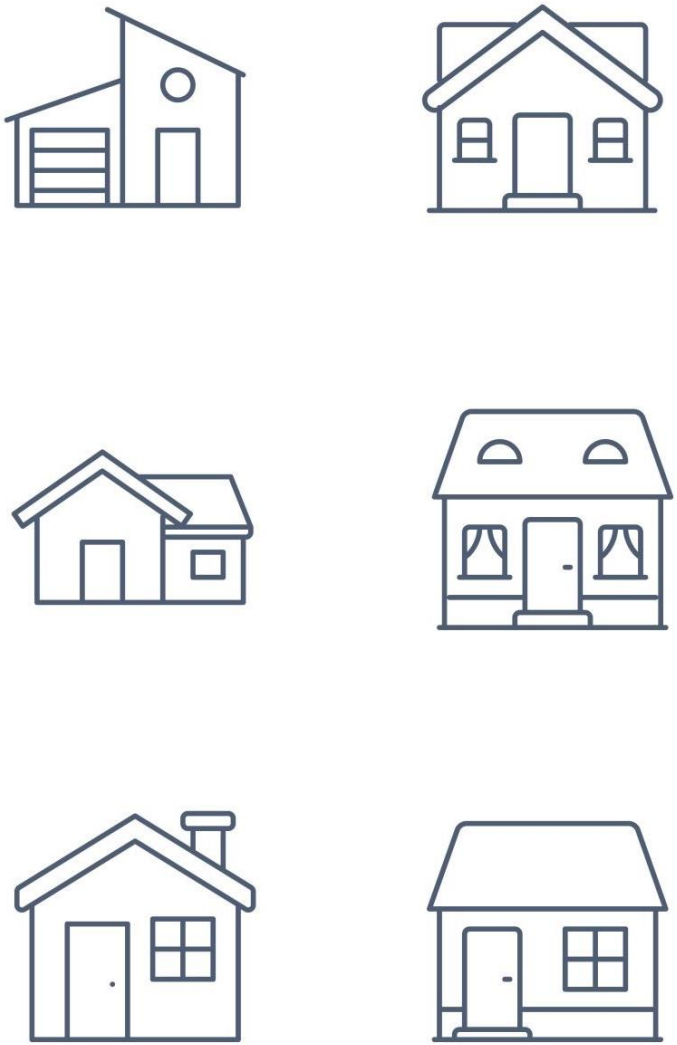
Key idea:

- If-else decision rules

What You Learned

- Linear Regression → predicts numbers
- Logistic Regression → predicts categories
- Decision Trees → rule-based decision system

Each model solves different types of problems



Review + Resources

GitHub materials:
https://github.com/AIWAZIRI-LABS/Introduction-to-AI-Machine-Learning-2026-ITng

Coding Examples

Day 7: Evaluation + Feature Engineering

ACCURACY, PRECISION, RECALL

ENCODING & SCALING

CONFUSION MATRIX

Making Machine Learning Models Better

Why this matters

Helps measure model performance

Improves accuracy and reliability

Prepares data for better learning

Core Evaluation Metrics

Accuracy

Percentage of correct predictions

Good for balanced datasets

Precision

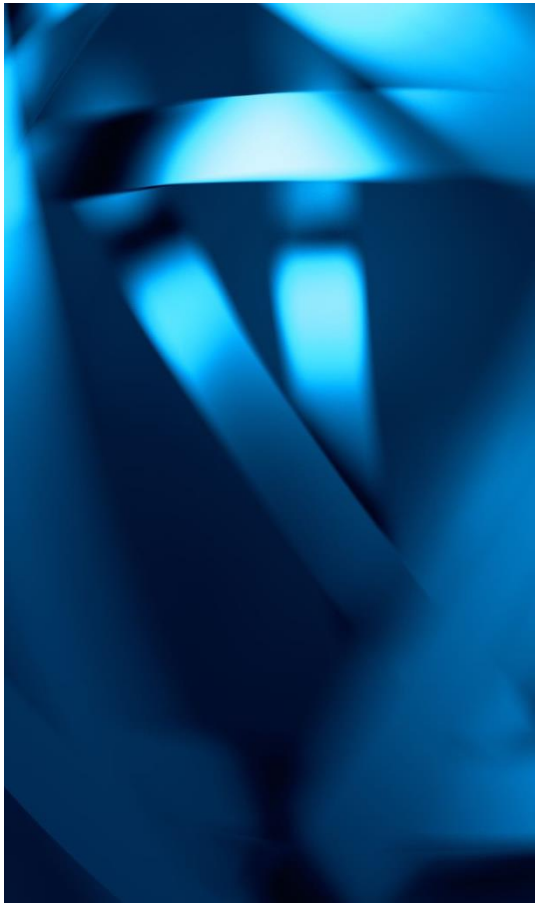
How many predicted positives are actually correct

Recall

How many actual positives were correctly identified

Key idea:

Accuracy alone is not always enough



Confusion Matrix: Understanding Prediction Results

What it is:

- A table that shows model performance:
- True Positive (TP)
- True Negative (TN)
- False Positive (FP)
- False Negative (FN)

Why it matters:

- Shows where the model is making mistakes
- Helps improve model performance

Converting Text into Numbers: Encoding

Why encoding is needed:

- ML models cannot understand text directly

Types:

- Label Encoding (assign numbers)
- One-Hot Encoding (create binary columns)

Example:

- Male = 0, Female = 1

Normalizing Data Values

What is scaling?

- Adjusting values to same range

Why it matters:

- Prevents large values from dominating
- Improves model performance

Common methods:

- Standardization (mean = 0, std = 1)
- Min-Max scaling (0 to 1 range)

What You Learned

ACCURACY IS NOT ENOUGH ALONE

PRECISION AND RECALL GIVE DEEPER INSIGHT

CONFUSION MATRIX SHOWS PREDICTION ERRORS

ENCODING CONVERTS TEXT TO NUMBERS

SCALING IMPROVES MODEL PERFORMANCE

Review + Resources

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Coding Examples

Thank you